

A recent MarketsandMarkets report projects the growth of the humanoid robot market from \$2.03 billion in 2024 to \$13.25 billion in 2029, with a compound annual growth rate (CAGR) of 45.5%. This growth will likely be driven by advances in AI and ML, improvements in humanoid robots, and the growing use of humanoids as educational robots. High initial investment and safety concerns need to be addressed to catalyse the growth in 2025 further.

Sustainability and green initiatives

The amount of computing power required for AI implementations, the data centres populating the cloud, and the uncountable number of mobile devices at the edge—the carbon footprint of all this might even be a bit too much for AI to comprehend. Still, deep within our hearts, we all know that we are heading towards destruction! The environmental impact does not end with the energy consumed by these, but also the resources consumed and the pollution caused during their production. Understandably, the tech industry and its large enterprise consumers are under great pressure from global agencies and governments to reduce and reverse their environmental impacts.

The industry is once again turning to its trusted friend—technology itself—to find ways to reduce energy consumption and improve sustainability by optimising architectures, hardware, code, and algorithms. They are also turning to renewable energy sources like solar energy and opting for more eco-friendly and sustainable manufacturing methods and packaging.

Apart from the actual tech processes, companies also need to optimise the carbon footprint of running the organisation per se, like transport and energy consumption on



AI-powered robots are expected to make their presence felt in the physical world

campus. So, companies have started using electric vehicles (EVs) in their fleet, and incentivising eco-friendly activities like using the stairs, avoiding plastic containers, and so on. While implementing these will be expensive initially, companies can no longer escape the regulatory requirements. That said, there will be significant cost savings in the long run.

Almost all industry majors, including Apple, Google, and Microsoft, have committed to becoming carbon neutral or negative by 2030. Carbon neutrality requires changes across the entire business—from manufacturing processes and supply chain to product lifecycle and business operations—to ensure zero climate impact.

So how does one become carbon-negative? It means you do stuff to reverse environmental damage, like planting trees, generating enough renewable energy to feed back to the grid, capturing greenhouse gases from the environment, promoting the maintenance of rangelands, which sequester carbon dioxide in the soil and biomass, landfill management, and so on.

Companies can also gain carbon credits by funding organisations that engage in such activities; it is believed

that this can be a means to uplift tribal populations. Companies pay them handsomely to do what comes naturally to them—fostering the environment. Companies earn carbon credits, while the tribes earn their sustenance with due respect!

And some more...

Spatial computing: Spatial computing is a paradigm that digitally enhances the physical world. An amalgam of various technologies like VR, augmented reality (AR), and AI, it finds applications in brand marketing, digital twins, and so on. For example, with spatial computing, you can just look at an object and seek information about it—an implied visual search. This technology has been given many names, discussed, hyped, and shelved many times in the past couple of years. However, with improvements in AI, ubiquitous connectivity, easy access to computing power, and availability of more comfortable headsets, analysts have again proposed a re-emergence of spatial computing this year—maybe not in the big, bad world but in smaller, closed spaces!

Telcos' trust with AI: The telco industry is expected to tap into the benefits of agentic AI for tasks like suggesting personalised and optimised